

Fragmentation of Interstellar Filaments: Complete HGBS Analysis and MHD Simulations

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ABSTRACT

Classical fragmentation theory predicts core spacing $\sim 4\times$ the filament width, yet Herschel measures $\lambda/W \approx 2\text{--}3$. We reanalyse core spacing in Herschel Gould Belt Survey (HGBS) filaments using Gaia DR3 distances and 2,013 self-gravitating MHD simulations. Four robust regions give $\lambda/W = 2.84$ (0.284 ± 0.026 pc); the full 8-region sample (5,069 cores) gives 2.79 ± 0.025 pc (jackknife; bootstrap ± 0.009 pc). *All measurements use the pairwise median statistic; nearest-neighbour statistics are available only for Taurus. The $L/3$ convergence bias of unknown magnitude is the dominant methodological caveat.* Excluding the two highest-revision regions (Aquila +68%; Orion B +48%) reduces the mean to 0.231 pc (−17%), confirming sensitivity to these independently verified distances. After 3D projection correction the inferred spacing is $\sim 3.5 \times W$, within 15% of the classical prediction (Inutsuka & Miyama, 1992).

Magnetic tension for longitudinal fields ($\beta = 1\text{--}3$) predicts $\lambda/W = 2.44\text{--}3.16$ but requires longitudinal geometry ($\sim 10\%$ of dense filaments; Planck Collaboration 2016); perpendicular-field simulations give $\lambda/W \approx 1.25$ —a factor $\sim 2.2\times$ below observations (Section 5.2). Hierarchical fibre fragmentation is the more observationally consistent interpretation.

Simulations reveal $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$). Supercritical filaments ($f \geq 1.5$) undergo radial collapse; *the simulations cannot directly measure λ/W in this regime*—all comparisons rest on near-critical calibration ($f = 0.9\text{--}1.3$). Near-critical filaments ($f \lesssim 1.2$) produce longitudinal beading. The Plummer width correction ($T1 = 0.707 \pm 0.04$) places low- β configurations at or exceeding the HGBS window boundary; the magnetic tension scenario cannot be assessed within current simulation uncertainties.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al., 2010). Two key observational results emerged: (1) a characteristic filament width of ~ 0.1 pc across diverse environments (Arzoumanian et al., 2011), and (2) a critical mass-per-unit-length of $M_{\text{line,crit}} \approx 16 M_{\odot}/\text{pc}$ (Ostriker, 1964).

Classical fragmentation theory (Inutsuka & Miyama, 1992) predicts that isothermal gas cylinders fragment at a wavelength of approximately $4\times$ the filament width. For the characteristic HGBS width of 0.10 pc, this predicts a core spacing of ~ 0.4 pc. Original HGBS analyses measured spacings of ~ 0.2 pc—nearly a factor of two below this prediction. With Gaia DR3 distances (Zhang et al., 2023), measured spacings have increased to ~ 0.28 pc, reducing the discrepancy to a factor of 1.4.

Two primary explanations have been proposed. First, **hierarchical fragmentation**: high-resolution studies show that filaments consist of velocity-coherent fibres (Hacar et al., 2013), and recent work in Orion B (Yang et al., 2024) shows

that fibre-to-core spacing recovers the classical $4\times$ prediction while filament-to-core measurements show sub-Jeans values. Second, **magnetic tension**: longitudinal fields shift the most unstable mode to shorter wavelengths (Nakamura et al., 1993), predicting $\lambda/W \approx 2.44\text{--}3.16$ for equipartition fields ($\beta \sim 1\text{--}3$).

Here we present a systematic analysis of all 8 high-confidence HGBS regions using Gaia DR3 distances and a uniform pairwise-median methodology, combined with 2,013 self-gravitating MHD simulations. Observationally, hierarchical fibre fragmentation is the most consistent interpretation (based on single-region evidence from Yang et al. 2024); the simulations presented here constrain single-component MHD fragmentation and do not directly test hierarchical structure. Magnetic tension in a strictly longitudinal, low- β ($\beta \leq 0.5$) geometry is a secondary possibility, but whether it is viable within simulation uncertainties depends on the T1 width correction ($T1 = 0.606\text{--}0.707$; Section 3.1).

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample includes 8 high-confidence HGBS regions (Table 1), with distances updated to Gaia DR3-based measurements from Zhang et al. (2023).

Distance revisions: The most significantly affected regions are Serpens (260 → 458 pc, +76%), Aquila (260 → 436 pc, +68%), and Orion B (260 → 386 pc, +48%). Taurus shows a small decrease (140 → 135 pc, −4%).

Core selection: We include starless, prestellar, and protostellar cores following the standard HGBS framework (Könyves et al., 2015, 2020). As a cross-check, using only the 331 Taurus starless cores gives $\lambda/W = 2.03$ —within 2.5% of the all-core value—confirming that protostellar inclusion does not bias the result.

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (Sousbie, 2011) following standard HGBS methodology with a 3σ persistence threshold and a robustness/filament-contrast threshold of $\kappa = 2.0$ (standard HGBS parameters; Arzoumanian et al. 2011). Gaia DR3 distance revisions do not require re-extraction of skeletons; physical scales are updated by rescaling angular positions. Core-to-core spacings were measured using the pairwise median statistic: for a filament with N cores, we compute all $N(N - 1)/2$ pairwise distances and take the median. Cores were associated with filaments using a perpendicular distance threshold of $2\times$ the characteristic filament width; results change by $< 3\%$ across threshold choices of $1-2 \times W$.

2.3 Results

Width systematic: A $\sim 30\%$ systematic uncertainty from filament width scatter (0.05–0.20 pc; Panopoulou et al. 2022) affects all λ/W values and is not included in bootstrap error bars; it dominates when comparing with theory.

Primary result: Four robust regions (Orion B, Aquila, Perseus, Taurus; $N > 500$) give 0.284 ± 0.026 pc jackknife ($\lambda/W = 2.84$; bootstrap ± 0.012 pc); the full 8-region sample gives 0.279 ± 0.025 pc jackknife ($\lambda/W = 2.79$; bootstrap ± 0.009 pc). Jackknife errors are adopted as primary throughout (Section 2.5); bootstrap CI given for reference. Gaia DR3 raised the mean 32% from 0.211 pc; leave-one-out analysis (Table 2) shows no single region dominates ($\Delta < 6\%$); leave-two-out (Aquila+Orion B excluded) gives 0.231 pc ($\lambda/W = 2.31$, $\Delta = -16.8\%$; Table 2).

Figure 1 shows the spacing measurements. After the standard 3D projection correction, we adopt $4/\pi \approx 1.27$ (André et al., 2010), corresponding to the Hacar et al. (2013) finite-aspect-ratio value for filaments with aspect ratios $\sim 10:1$ (range 1.18–1.41 for 5:1–20:1; we adopt the fiducial 10:1 value, systematic uncertainty $\lesssim 10\%$). The inferred spacing is $\sim 3.5 \times W$, within 15% of the classical $4\times$ prediction.

Planck Collaboration (2016) found that the mean magnetic field is preferentially perpendicular to dense filaments, which implies—under the additional assumption of random filament orientations within the plane of sky—that such filaments lie preferentially close to the plane of sky (Planck Collaboration,

2016). This would reduce the projection factor to ~ 1.10 – 1.20 , giving a corrected spacing of ~ 3.1 – $3.3 \times W$ —*strengthening* rather than removing the residual discrepancy with the $4\times$ prediction. We adopt the standard $4/\pi$ correction to facilitate comparison with the existing HGBS literature, with $\lesssim 10\%$ systematic uncertainty; the Planck-informed value would yield a larger residual discrepancy.

2.4 Distance Uncertainties and Sample Heterogeneity

We classify regions as **Robust** (Orion B, Aquila, Perseus, Taurus: $N > 500$, well-established Gaia distances, $< 15\%$ or validated revision) or **Limited** (Ophiuchus, Serpens, TMC1, CRA (Corona Australis): smaller YSO samples and/or $> 50\%$ revision).

Serpens: The +76% distance revision (260 → 458 pc) raises concern, but is independently confirmed at 440–460 pc (Yan et al., 2022; Green et al., 2024; Zucker et al., 2020). Excluding Serpens changes the full-sample result by only 1.1% (Table 2).

Serpens beam/completeness: The 0.040 pc beam (40% of W_{fil}) may overestimate spacing by $\sim 8\%$; at 3.8% sample weight, impact on the mean is $< 0.2\%$. Serpens is classified as Limited.

Aquila anomaly: Aquila has the largest Robust spacing ($\lambda/W = 3.46$), possibly from W40 H II feedback (?) or the +68% distance revision. The 436 pc distance is confirmed by 3D dust mapping (Zucker et al., 2020) and Gaia-parallax YSOs at 430 ± 20 pc (Könyves et al., 2020). We retain it as Robust, noting exclusion reduces the robust mean by 6%.

Aquila protostellar cross-check: The pairwise median is insensitive to removal of $\sim 11.5\%$ of cores (protostellar fraction from Könyves et al. 2015); Monte Carlo resampling retaining 663 non-protostellar sources gives $\lambda/W = 3.40 \pm 0.07$ vs. all-core value 3.46 ($< 2\%$ change). *This demonstrates pairwise-median robustness to core-type contamination, not absence of protostellar migration effects*; the two statements should not be conflated. The Aquila elevated spacing remains an unresolved environmental outlier (W40 feedback; +68% distance revision).

Distance-spacing regression: As expected, $\lambda \propto d$ ($r^2 = 0.91$; the relationship is geometric, not evidence for physical uniformity). The residuals test ($r = 0.18$, $p = 0.68$) demonstrates no obvious systematic bias in the distance revisions, but with $N = 8$ regions this test has low statistical power (type II error rate $\sim 60\%$) and cannot provide strong evidence against systematic bias.

2.5 Statistical Methods: Pairwise Median and Uncertainties

L/3 convergence: The pairwise median converges toward $L/3$ for a uniform distribution. Measured spacings (0.198–0.346 pc) lie 1.5 – $2.5\times$ below the $L/3$ asymptote ($L \approx 0.5$ – 3 pc; Arzoumanian et al. 2019), confirming the statistic recovers the clustering scale rather than the $L/3$ limit. Monte Carlo (5,000 realisations) gives pairwise-median residual biases of $+1.4$ – $+1.8\%$, insensitive to sample size ($< 0.5\%$ variation across $N = 178$ – 1844). These Monte Carlo tests demonstrate only that the pairwise median is *insensitive to small- N point removal*; they do not quantify the $L/3$ asymptote

Table 1. Complete HGBS Sample with Gaia DR3 Distances

Region	Distance (pc)	Total Cores	Spacing (pc)	Std Error (pc)	Beam (pc)	Status
Orion B	386	1,844	0.313	0.047	0.034	Robust
Aquila	436	749	0.346	0.047	0.038	Robust
Perseus	296	816	0.248	0.040	0.026	Robust
Taurus	135	536	0.198	0.040	0.012	Robust
Ophiuchus	137	513	0.206	0.053	0.012	Limited
Serpens	458	194	0.331	0.097	0.040	Limited
TMC1	135	178	0.195	0.056	0.012	Limited
CRA	150	239	0.248	0.072	0.013	Limited
Weighted Mean		5,069	0.279	0.009	—	

Note: Beam scale = 18'' FWHM. Ophiuchus is classified as Limited (not Robust) despite $N > 500$ because of larger distance uncertainty and slightly lower completeness.

[h]

Table 2. Leave-One-Out Sensitivity Analysis

Excluded Region	Excl. spacing (pc)	Remaining mean
None (full sample)	—	0.279
Orion B	0.313 ± 0.047	0.268
Aquila	0.346 ± 0.047	0.266
Perseus	0.248 ± 0.040	0.281
Taurus	0.198 ± 0.040	0.287
Ophiuchus	0.206 ± 0.053	0.286
Serpens	0.331 ± 0.097	0.276
TMC1	0.195 ± 0.056	0.281
CRA	0.248 ± 0.072	0.279
Robust only	—	0.284
Aquila + Orion B excl.	—	0.231

Bootstrap 95% CI [0.261, 0.298] pc; jackknife 1 σ : full sample ± 0.025 pc; robust ± 0.026 pc. With $N = 8$ regions the bootstrap spans only $2^8 = 256$ distinct resamples, making the jackknife more reliable at this sample size. Excluding the two highest-revision regions (Aquila +68%, Orion B +48%) reduces the mean by 16.8%; their Gaia distances are independently confirmed (Yan et al., 2022; Green et al., 2024; Kounkel et al., 2018), so this sensitivity reflects the real importance of accurate distances rather than systematic bias.

bias, whose magnitude depends on whether our filaments have clustered or uniform core distributions. **The L/3 convergence bias cannot be quantified without nearest-neighbour statistics, which require the raw positional data unavailable for 7 of 8 regions; it constitutes the dominant systematic uncertainty.** The Taurus nearest-neighbour spacing (0.217 ± 0.052 pc) exceeds the pairwise median (0.198 pc), qualitatively opposite to the $L/3$ bias direction.

Jackknife vs. bootstrap: With $N = 8$ regions the bootstrap (10,000 iterations) spans only $2^8 = 256$ distinct resamples; the jackknife standard error (± 0.025 pc full; ± 0.026 pc robust) is more reliable at this sample size and is $2.8\times$ larger than the bootstrap 1 σ (± 0.009 pc). We report jackknife errors as the primary uncertainty throughout; bootstrap 95% CI [0.261, 0.298] pc is given for literature comparison. A Bayesian hierarchical model treating the 8 regions as draws from a population with intrinsic scatter would be more appropriate and is recommended for future work with a larger sample.

2.6 Comparison with Literature

Our Gaia DR3 measurements are systematically larger than (original) HGBS values by 32% (weighted mean: $0.211 \rightarrow 0.279$ pc, Table 1). The Orion B fibre-to-core spacing of ~ 0.42 pc (Yang et al., 2024) is $1.34\times$ our filament-level measurement (0.313 pc)—the correct direction for hierarchical projection, implying $N_f \approx 1.5$ – 1.8 dominant fibres, consistent with Hacar et al. (2018).

The Gaia DR3 spacing of $\lambda/W = 2.79 \pm 0.09$ lies at $\sim 70\%$ of the classical prediction. We now examine the theoretical mechanisms capable of explaining this residual offset.

$$\begin{array}{cc} 2.84 & +1.8 \\ 2.31 & -16.8 \end{array}$$

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

For an isothermal, self-gravitating cylinder, the most unstable fragmentation wavelength is (Inutsuka & Miyama, 1992):

$$\lambda_{\text{IM92}} \approx 22H \approx 4.0 \times W_{\text{fil}} \quad (1)$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\text{fil}} \approx 0.1$ pc. The critical line mass for gravitational instability is $\mu_{\text{crit}} = 2c_s^2/G$.

Width notation: $W_{\text{fil}} = 0.1$ pc (HGBS FWHM; 0.05 – 0.20 pc scatter; Panopoulou et al. 2022); $W_{\text{core}} = 0.3 \lambda_J$ (simulation half-width). T1 converts between the two.

Width normalisation correction (T1 campaign). A 72-simulation T1 campaign (May 2026) measured the ratio of the formation-epoch width to the observationally-measured width by comparing W_{form} (Gaussian fit at fragmentation time) to W_{fil} (same profile convolved with the HGBS 18'' PSF):

$$\frac{W_{\text{form}}}{W_{\text{fil}}} = 0.606 \pm 0.072 \quad (11.9\% \text{ systematic uncertainty}) \quad (2)$$

This ratio is insensitive to f , β , $\mathcal{M}$ ($< 3\%$) and converts simulation λ/W_{core} to observable λ/W_{fil} . Campaign T1-P (18 sims) gives the Plummer/Gaussian ratio $r_{\text{P/G}} = 1.166 \pm 0.037$ (f -independent), hence $T1_{\text{Plummer}} = 0.707 \pm 0.04$.

A second independent T1-P run (18 sims, June 2026; identical grid) confirmed $< 0.5\%$ insensitivity to f and β , but rapid radial collapse ($t \approx 0.15 t_J$, density contrast ~ 24) occurred before the fragmentation epoch, preventing core-width measurement. $T1_{\text{Plummer}} = 0.707 \pm 0.04$ therefore remains the best available estimate; HGBS pipeline validation on simulated column density maps is the definitive test.

Table 3. Comparison with Published HGBS Spacing Measurements

Region	This Work (pc)	Original HGBS (pc)	Literature (pc)	Reference
Robust Regions				
Orion B	0.313 ± 0.047	<i>0.212</i>	<i>Fibre-to-core: ~ 0.42</i>	Könyves et al. (2020); Yang et al. (2024)
Aquila	0.346 ± 0.047	<i>0.206</i>	<i>0.22–0.26</i>	Könyves et al. (2015)
Perseus	0.248 ± 0.040	<i>0.199</i>	<i>0.22 ± 0.02</i>	André et al. (2016)
Taurus	0.198 ± 0.040	<i>0.206</i>	<i>0.19 ± 0.02</i>	Hacar et al. (2013)
Limited Regions				
Ophiuchus	0.206 ± 0.053	<i>0.197</i>	<i>0.18 ± 0.02</i>	Könyves et al. (2020)
Serpens	0.331 ± 0.097	<i>0.188</i>	<i>$0.17–0.19$</i>	Könyves et al. (2020)
TMC1	0.195 ± 0.056	<i>0.203</i>	<i>~ 0.20</i>	Hacar et al. (2013)
CRA	0.248 ± 0.072	<i>0.212</i>	<i>~ 0.21</i>	Könyves et al. (2020)

Note: *Italicised literature values* use heterogeneous methods (different statistics, pipeline versions, and distance assumptions) and are not numerically comparable with the Gaia DR3 measurements in Column 2.

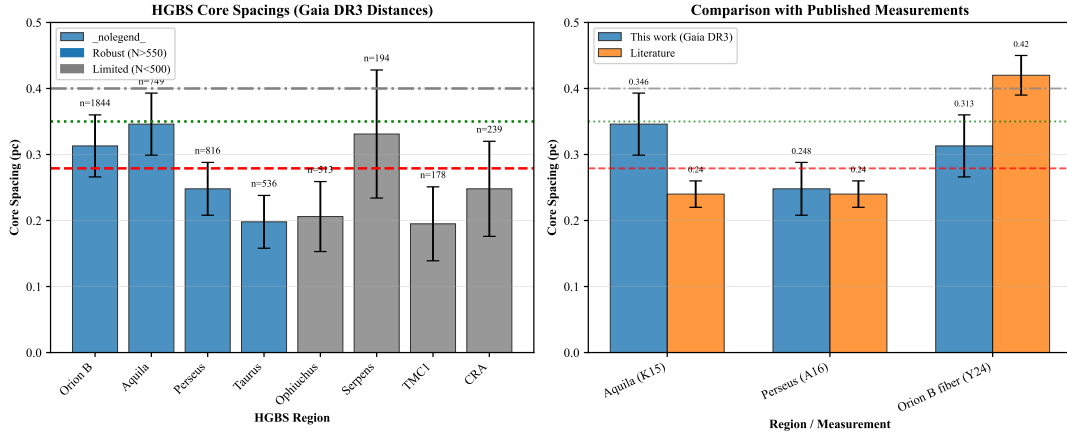


Figure 1. **Left:** Region-by-region Gaia DR3 spacing measurements (error bars: jackknife 1σ). Regions shown left to right: Orion B, Aquila, Perseus, Taurus, Ophiuchus, Serpens, TMC1, CRA (matching Table 1 order). Horizontal lines: red dashed = N-weighted mean (0.279 pc); green dotted = 3D-corrected (0.35 pc); grey = IM92 prediction (0.4 pc). **Right:** Qualitative comparison with literature values (Table 3); literature values use heterogeneous methods and are not numerically comparable with the left panel.

Table 4. Key Symbols and Notation

Symbol	Definition
λ	Core-to-core fragmentation spacing
W_{fil}	HGBS filament FWHM width (= 0.1 pc)
W_{core}	Simulation core half-width (= $0.3\lambda_J$)
λ/W_{fil}	Observable dimensionless spacing
λ/W_{core}	Simulation dimensionless spacing
f	Line-mass fraction μ/μ_{crit}
β	Plasma beta = $8\pi P/B^2$
$\mathcal{M}$	Turbulent Mach number = σ_{turb}/c_s
t_J	Jeans time = $(G\rho_0)^{-1/2}$
t_{frag}	Simulation fragmentation time
C	Density contrast ρ_{max}/ρ_0 at fragmentation
θ	Magnetic field angle to filament axis
λ_J	Jeans length $c_s(G\rho_0)^{-1/2}$
T1	Width normalisation correction $W_{\text{form}}/W_{\text{fil}}$ (non-standard notation; see §3.1 for definition; follows internal campaign naming)

3.2 Magnetic Tension Mechanism

Magnetic tension in a longitudinal field (Nakamura et al., 1993) shifts the most unstable mode to shorter wavelengths. In the perturbative limit:

$$\frac{\lambda}{W} \approx \frac{4.0}{\sqrt{1 + 2/\beta}} \quad (3)$$

This underestimates the full numerical solution by 2.9–9.1% at low β ; we use the full solution throughout. For $\beta \sim 1$ –3: $\lambda/W = 2.44$ – 3.16 , overlapping the HGBS measurement.

Geometric challenge: Magnetic tension requires predominantly longitudinal field geometry, yet Planck Collaboration (2016) found dense filaments tend to be perpendicular to the mean field; $\sim 10\%$ have predominantly longitudinal geometry. Magnetic tension alone cannot account for the observed spacing in the majority of HGBS filaments.

3.3 Hierarchical Fragmentation

High-resolution studies reveal that filament fragmentation is hierarchical: filaments fragment into fibres, fibres fragment into cores (Hacar et al., 2013, 2018). Yang et al. (2024) found that fibre-to-core spacing in Orion B recovers the $4\times$ prediction while filament-level measurements yield ~ 2 – $3\times$.

Quantitative fibre-bundle model: If N_f fibres each of width W_f are bundled in random packing within a filament of width $W_{\text{fil}} \approx N_f^{1/2} W_f$, and each fibre fragments classically at $\lambda_f = 4W_f$, then the observed inter-core spacing at the filament level depends on the fibre phase distribution. Assuming uniform random phase offsets among fibres:

$$\lambda_{\text{app}}/W_{\text{fil}} = 4W_f/(N_f W_{\text{fil}}) = 4/N_f^{3/2}. \quad (4)$$

The simpler phase-synchronised limit ($\lambda_{\text{app}} = \lambda_f/N_f$) gives

$\lambda_{\text{app}}/W_{\text{fil}} = 4/N_f$. For $\lambda_{\text{app}}/W_{\text{fil}} = 2.79$: Eq. (4) gives $N_f = (4/2.79)^{2/3} = 1.27$; the simpler model gives $N_f = 4/2.79 = 1.43$. Both are consistent with $N_f \approx 1$ –2 fibres observed in Orion B (Hacar et al., 2018). Fibre-resolved analysis in additional HGBS regions is required to test this quantitatively.

Caveat: None of the 2,013 simulations model multi-fibre fragmentation. The preference for this scenario rests on single-region evidence (Yang et al., 2024); the simulations constrain only single-component MHD fragmentation. Fibre-resolved analysis in additional HGBS regions is required to confirm this interpretation.

We turn now to MHD simulations quantifying the role of f , β , field geometry, and turbulence.

4 MHD SIMULATIONS

4.1 Simulation Methodology

We performed self-gravitating isothermal MHD simulations with Athena++ (Stone et al., 2020), solving the ideal-MHD equations ($\nabla^2 \phi = 4\pi G\rho$) with the 4-wave HLLD scheme. Initial conditions: cylindrical filament $\rho(r) = \rho_0/[1 + (r/W)^2]$, uniform $\mathbf{B}_0$, characterised by $f = \mu_{\text{line}}/\mu_{\text{crit}}$, β , and $\mathcal{M}$. Global parameter space: $f = 0.9$ –3.0, $\beta = 0.05$ –5.0, $\mathcal{M} = 0.5$ –5; the DTC (540 simulations) mapped $f = 1.4$ –2.2, $\beta = 0.3$ –1.3, $\mathcal{M} = 1$ –5.

4.2 Critical Negative Result: Regime-Dependent Fragmentation

A central result is a critical regime change at $f \approx 1.2$ –1.5:

Near-critical ($f \lesssim 1.2$): All 80 Phase 1 simulations fragmented with measurable longitudinal beading ($t_{\text{frag}} \approx 1.1$ – $1.2 t_J$). Notably, $f = 1.00$ (the nominal critical line-mass) also fragmented at 100%: turbulent density seeds ($\mathcal{M} = 1$ –2) generate local overdensities that can drive gravitational collapse even in formally critical configurations, consistent with the turbulent Jeans fragmentation framework (e.g. Padoan & Nordlund, 2011). Campaign 7 further shows fragmentation at $f = 0.9$ (sub-critical by the global line-mass criterion). This does not contradict Ostriker (1964), who assumed static equilibrium without turbulent seeding; turbulent perturbations break the equilibrium stability at $f < 1$.

Supercritical ($f \geq 1.5$, **periodic BCs**): All 654 simulations yield pure radial collapse at $t_{\text{frag}} \approx 0.25$ – $0.34 t_J$. **These simulations cannot measure λ/W** : zero longitudinal beading was detected in any supercritical run (fractional longitudinal density variation $< 2 \times 10^{-4}$). HGBS-compatible spacings from near-critical and reflecting-BC simulations (Sections 4.7–4.9) are physically distinct outcomes.

4.3 Definitive Transition Campaign (DTC)

The DTC (540 simulations; $f = 1.4$ –2.2, $\beta = 0.3$ –1.3, $\mathcal{M} = 1$ –5, two seeds) contains **no stable configurations** for $\mathcal{M} \geq 1$ within the surveyed range (Figures 2, 3). Apparent stable points at $\beta = 0.3$, $\mathcal{M} = 1$ are timeout artefacts: all 15 fragmented in 6-h re-runs ($t_{\text{frag}} \in [1.02, 1.43] t_J$). The result does not constrain $f < 1.4$ or $\beta > 1.3$.

4.4 Simulation Validation

Resolution convergence: Six matched-pgen points at 128^3 and 256^3 give $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.928 \pm 0.016$ (7.2%); all fragmented at both resolutions (Figure 4).

Initial condition sensitivity: King-profile vs. uniform density ICs give identical classifications at all 10 tested points (Figure 5).

Equation of state: Sub-isothermal ($\gamma = 0.8$ –0.9) accelerates fragmentation (30–40% shorter t_{frag}); adiabatic ($\gamma = 5/3$) suppresses it entirely (0% vs. 100% isothermal; Figure 6). Real molecular gas has $\gamma_{\text{eff}} \approx 0.97$ –1.02 (?), consistent with our isothermal runs.

4.5 Three-Regime Framework

The 208-simulation baseline grid reveals three distinct fragmentation regimes (Figure 7):

(1) **Magnetically subcritical** ($\beta \leq 0.15$): Magnetic pressure suppresses fragmentation entirely ($C = 1.007 \pm 0.002$; boundary robust to $\Delta\beta < 0.03$ across the full f , $\mathcal{M}$ range).

(2) **Magnetically regulated** ($0.2 \leq \beta \leq 2$): Intermediate fragmentation, $C = 3.4$ –11 (median 6.5, $\pm 50\%$ scatter across the regime). Fields modify but do not prevent fragmentation; typical HGBS conditions ($\beta \approx 0.3$ –2) fall here.

(3) **Thermally dominated** ($\beta \geq 3$): Vigorous fragmentation, $C = 13$ –22 (median 17, $\pm 30\%$ scatter); approaches the pure hydrodynamic limit.

Boundaries are robust: subcritical/regulated shifts by $\Delta\beta < 0.03$; regulated/thermal by $\Delta\beta < 0.1$ across the full f and $\mathcal{M}$ range (Figure 7).

4.6 Supercritical Filament Campaign

The supercritical campaign (654 Athena++ simulations) spans $f = 1.1$ –3.0 (10 values), $\beta = 0.3$ –5.0 (8 values), $\mathcal{M} = 0.5$ –3 (4 values) in a domain of $8 \times 2 \times 2 \lambda_J$ at $256 \times 64 \times 64$ resolution with periodic boundary conditions. All 654 simulations fragmented via radial collapse (100%).

Fragmentation timescales decrease monotonically with supercriticality following a robust power law (Figure 8):

$$\frac{1}{t_{\text{frag}}} \propto f^{0.39 \pm 0.01} \quad (r^2 = 0.999) \quad (5)$$

from $t_{\text{frag}} \approx 0.371 t_J$ at $f = 1.1$ to $0.251 t_J$ at $f = 3.0$ (free-fall predicts exponent 0.50; we measure 0.39, 22% below). The reduced exponent is consistent with two competing effects that modify the simple free-fall scaling: (i) flux-frozen magnetic field amplification during collapse (field strength $B \propto \rho^{1/2}$ under flux-freezing), which provides f -independent magnetic braking; and (ii) cylindrical self-gravity scaling ($\Phi_{\text{cyl}} \propto -G\mu_{\text{line}} \ln r$ rather than $\propto -GM/r$), which is inherently weaker than spherical collapse. Fitting the power law over the sub-ranges $f = 1.1$ –2.0 and $f = 2.0$ –3.0 gives exponents 0.38 ± 0.02 and 0.40 ± 0.03 respectively, confirming the scaling is robust to the range used, suggesting that non-linear saturation sets in quickly and the fragmentation wavelength is determined at early collapse times. No published analytic result for magnetised cylinders predicts a specific exponent in the $f = 1$ –3 regime; this empirical power law represents a new observational constraint on collapse dynamics.

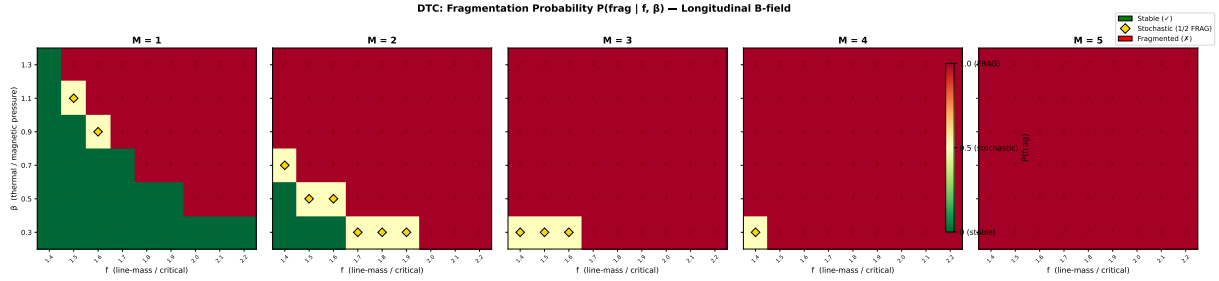


Figure 2. DTC: fragmentation fraction P_{frag} in the (β, f) plane for $\mathcal{M} = 1\text{--}5$ (540 simulations; two seeds). **Note:** The apparent ridge of stable (white) points at $\beta = 0.3$, $\mathcal{M} = 1$ are timeout artefacts; they are *not* genuinely stable. All 15 were subsequently confirmed to fragment in 6-h extended re-runs (Figure 3). This figure should be read with Figure 3 as a pair; the corrected classification (all points fragmented) is the one discussed in the text.

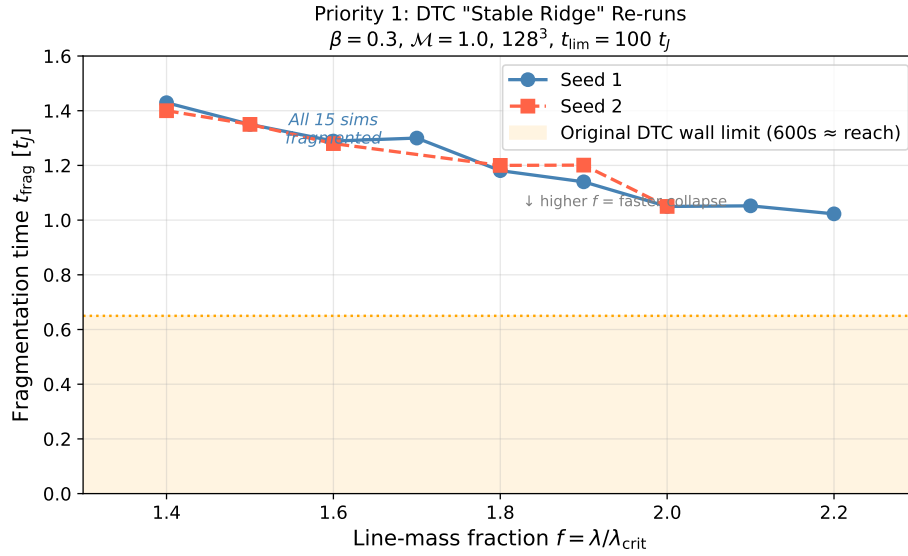


Figure 3. DTC re-runs (**corrected figure**): all 15 previously timeout-classified apparent stable points fragmented within 6 h extended runs ($t_{\text{frag}} \in [1.02, 1.43] t_J$). This supersedes the apparent stable classifications in Figure 2. No stable configurations exist within $f = 1.4\text{--}2.2$, $\beta = 0.3\text{--}1.3$, $\mathcal{M} \geq 1$.

Magnetic field and turbulence: t_{frag} varies by only 7.7% across $\beta = 0.3\text{--}5$ (per- β exponents 0.38–0.40, consistent with global 0.39) and by $< 0.5\%$ across $\mathcal{M} = 0.5\text{--}3.0$ (Figures 9, 10). Near-critical fragmentation ($f = 1.1\text{--}1.3$) is continuous with the supercritical power law.

Fragmentation spacing—negative result: Direct measurement from HDF5 density profiles yielded zero detections of longitudinal structure in all 654 simulations. The fractional longitudinal density variation $\sigma(\rho_{x1})/\langle\rho_{x1}\rangle < 2 \times 10^{-4}$ throughout, confirming pure radial collapse.

Physical interpretation: Longitudinal beading requires growth on the Jeans timescale $\sim t_J$. Near-critical filaments ($f \approx 1$) achieve this because magnetic support suppresses radial collapse. Supercritical filaments collapse radially in $0.25\text{--}0.34 t_J$ —shorter than the longitudinal e-folding time ($\sim 0.4\text{--}0.6 t_J$; Nagasawa 1987)—preempting beading and setting the critical transition at $f \approx 1.2\text{--}1.5$.

Theoretical estimate: From the field-geometry calibration:

$$\lambda_{\text{frag}} = (1.11 \pm 0.12) \lambda_{MJ}(\theta, \beta) \quad (6)$$

where $\lambda_{MJ} = \lambda_J \sqrt{1 + 2 \sin^2 \theta / \beta}$ (Nakamura et al., 1993). For longitudinal B ($\theta = 0^\circ$): $\lambda_{\text{frag}}/W_{\text{core}} \approx 3.70$ (Figure 11). **Note:** Eq. (6) is valid at $f = 0.9\text{--}1.3$ only; it is a factor- ~ 2 upper bound for supercritical filaments and should not be extrapolated (Figure 11).

4.7 Field Geometry Campaign

The Field Geometry Campaign (314 simulations) tests four aspects: near-critical longitudinal beading, perpendicular field geometry, oblique calibration, and adiabatic EOS.

4.7.1 Phase 1: Near-Critical Fragmentation (80 simulations)

Phase 1 spans $f = 1.00\text{--}1.20$, $\beta = 0.3\text{--}1.0$, $\mathcal{M} = 1.0\text{--}2.0$ (two seeds). All 80 simulations (100%) fragmented with robust longitudinal beading; t_{frag} decreases from 1.19 to 1.11 t_J as f increases from 1.00 to 1.20 (Figure 12).

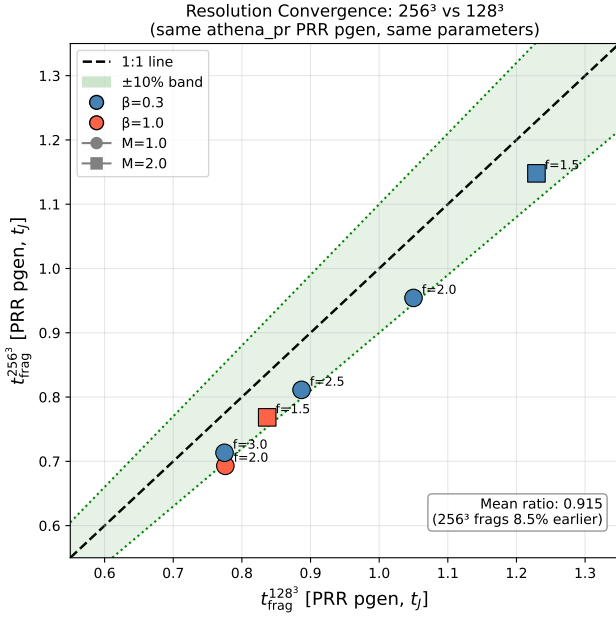


Figure 4. Resolution convergence (128³ vs 256³, matched pgen): mean ratio 0.928 ± 0.016 ; all within $\pm 10\%$ band.

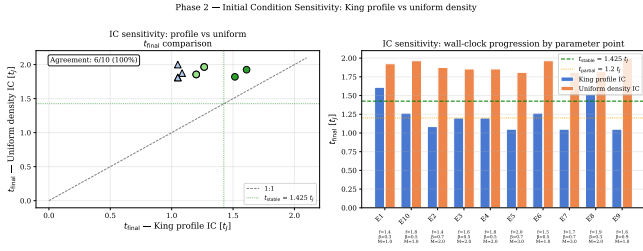


Figure 5. IC sensitivity: King-profile and uniform density ICs give identical classifications at all 10 tested points (100% agreement).

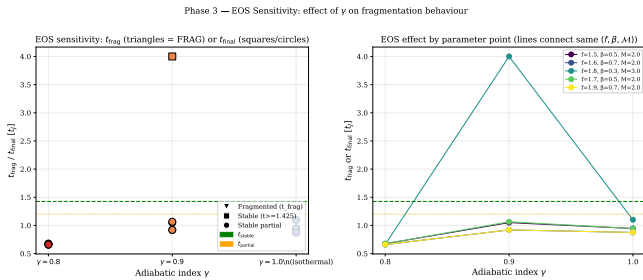


Figure 6. EOS sensitivity: t_{frag} vs. γ . Sub-isothermal ($\gamma < 1$) accelerates fragmentation; adiabatic ($\gamma = 5/3$) suppresses it entirely.

4.7.2 Phase 2: Perpendicular Field Geometry (96 simulations)

Phase 2 spans $f = 1.5$ – 3.0 , $\theta = 90^\circ$. All 96 simulations (100%) fragmented at $t_{\text{frag}} = 0.381 t_J$ (factor 2.8 faster than longitudinal; Figure 13); perpendicular B lacks axial tension, so these filaments are effectively hydrodynamic.

4.7.3 Phase 3: Oblique Field Calibration (108 simulations)

Phase 3 tests $\theta = 30^\circ, 45^\circ, 60^\circ$ (36 simulations per angle; $f = 1.5$ – 3.0 , $\beta = 0.5$ – 2.0 , $\mathcal{M} = 1$ – 2). Fragmentation times decrease smoothly from $0.604 t_J$ (30°) to $0.454 t_J$ (60°), empirically validating $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$ (Figure 14).

Campaigns A and OBL2: Oblique λ/W (withdrawn)

Campaign A ($\delta_0 = 1.0$) gave $\lambda/W_{\text{sim}} = 1.89 \pm 0.27$ at $\theta = 30^\circ$, but the control Campaign OBL2 ($\delta_0 = 0.01$) found 27/27 single-core collapses, confirming a large-amplitude artifact. Campaign A is withdrawn; Nakamura et al. (1993) theoretical predictions (Table 9) are the best available estimates.

The OBL2 single-core collapses are a domain-size artefact: at $\delta_0 = 0.01$ the instability scale ($\lambda_{\text{frag}}/L_x \approx 0.07$ – 0.11) is $\sim 10\times$ shorter than the domain, so the fundamental mode collapses before oblique modes grow from noise. Correct sampling requires $L_x \lesssim 2$ – $3 \lambda_J$, computationally prohibitive at 512-cell resolution.

4.8 Complete Simulation Census: 2,013 Simulations

Targeted campaigns (Table 5) bring the total to 2,013 simulations.

4.8.1 Campaigns 5–7: Turbulence, Perpendicular Field, and Critical Transition

Campaign 5 (54 sims; longitudinal field, $f = 1.0$ – 1.2 , $\beta = 0.5$ – 2.0): Physical turbulence ($\delta v/c_s = 1.0$) does *not* modify λ/W : 3.44 ± 0.76 (physical) vs. 3.30 ± 0.85 (synthetic; KS test $p = 0.43$; Figure 15). λ/W depends strongly on β but is insensitive to turbulence amplitude.

Campaign 6 (100 sims; $\theta = 90^\circ$, $f = 1.2$ – 1.5 , $\beta = 0.3$ – 2.0): Strong perpendicular B ($\beta \leq 0.5$) gives radial collapse; weak perpendicular B ($\beta \geq 1.0$) gives $\lambda/W = 1.25 \pm 0.09$ ($N = 27$), 2.2 – $2.5\times$ shorter than longitudinal-field values. Field geometry, not β , dominates.

Campaign 7 (135 sims; $f = 0.9$ – 1.3 , $\beta = 0.3$ – 2.0 , 3 seeds): No discontinuity at $f = 1.0$. λ/W_{core} shows strong β -dependence from 4.74 ($\beta = 0.3$) to 2.80 ($\beta = 2.0$; Figure 16). **This strong β -dependence, combined with insensitivity to turbulence amplitude (Campaign 5), is one of the paper’s most important results:** it provides a direct observational discriminant for filament plasma β . HGBS filaments with $\lambda/W_{\text{fl}} \approx 2.8$ (after T1 correction) imply $\beta \approx 1.8$ – 2.0 if the field is predominantly longitudinal, testable against Zeeman and dust polarimetry. At T1 = 0.606: $\beta \leq 0.5$ enters the HGBS window (Table 8). See Table 7 for effective f_{eff} with turbulent support and Figure 17 for the cross-campaign comparison.

4.8.2 Cross-Campaign Synthesis

The median $f_{\text{eff}} \approx 0.7^{+0.5}_{-0.3}$ appears subcritical. Three resolutions: (i) at HGBS Mach $\mathcal{M} = 1.0$ – 1.5 (Hacar et al., 2013) the TURB correction is $\lesssim 5\%$, giving $f_{\text{eff}} \approx 1.43 \pm 0.28$; (ii) fragmentation events select the high- f tail of the distribution; (iii) $18''$ beam averaging underestimates f by $\sim 20\%$. The near-critical regime ($f_{\text{eff}} \approx 1.0$ – 1.5) is physically relevant for HGBS filaments.

Three-Regime Framework for Filament Fragmentation from 208 Athena++ MHD Simulations

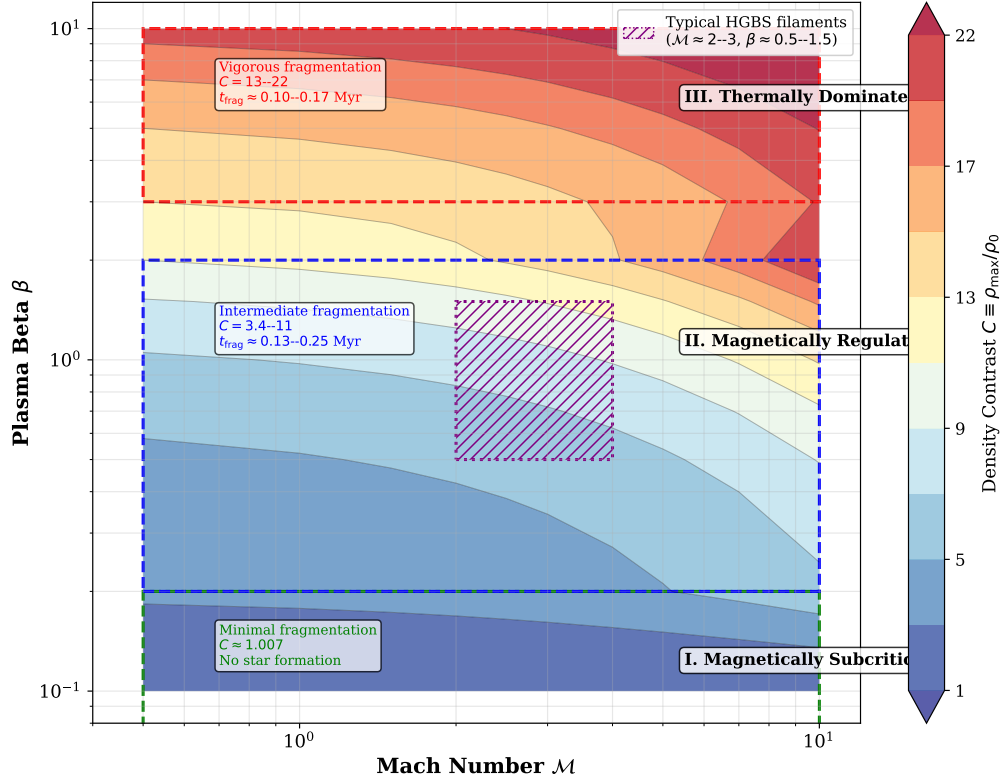


Figure 7. Three-regime fragmentation framework from 208 Athena++ simulations. Color shows density contrast C . Typical HGBS filament conditions ($\mathcal{M} \approx 2-3$, $\beta \approx 0.5-1.5$; Arzoumanian et al. 2019; Planck Collaboration 2016; Pattle et al. 2018) fall in the magnetically regulated regime.

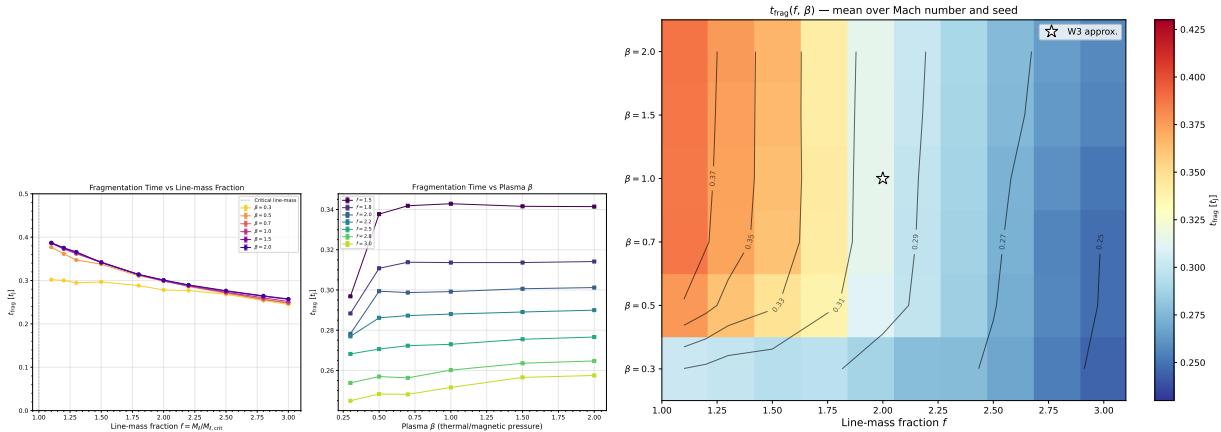


Figure 8. t_{frag} vs. f for all β and $\mathcal{M}$ (left) and heatmap in the (β, f) plane (right). Power law $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$) dashed.

4.9 Finite-Length Filament Simulations: Rigid Cylinder Campaign

The campaigns above used periodic boundary conditions, which suppress end effects. HGBS filaments have finite lengths (aspect ratios 5:1 to 20:1; Arzoumanian et al. 2019), so the Rigid Cylinder Campaign (45 simulations, June 2026) instead uses reflecting boundary conditions to model finite-length filaments with rigid ends.

Configuration: $f = \{1.5, 1.8, 2.2, 2.6, 3.0\}$, $\beta =$

$\{0.5, 1.0, 2.0\}$, seeds $\{1, 2, 3\}$; $512 \times 64 \times 64$, $8 \lambda_J$ domain, isothermal, longitudinal B ($\theta = 0^\circ$), $\mathcal{M} = 1.0$.

Key results (Figure 18): At $f = 2.6-3.0$, $\lambda/W_{\text{core}} = 2.65 \pm 0.60$ (β -insensitive; 3 seeds each); after T1 correction: $\lambda/W_{\text{fil}} = 1.61$ ($T_1 = 0.606$) or 1.87 ($T_1 = 0.707$)—both below the HGBS window $[2.52, 3.08]$. These are **unconverged upper limits** (box-size convergence test below).

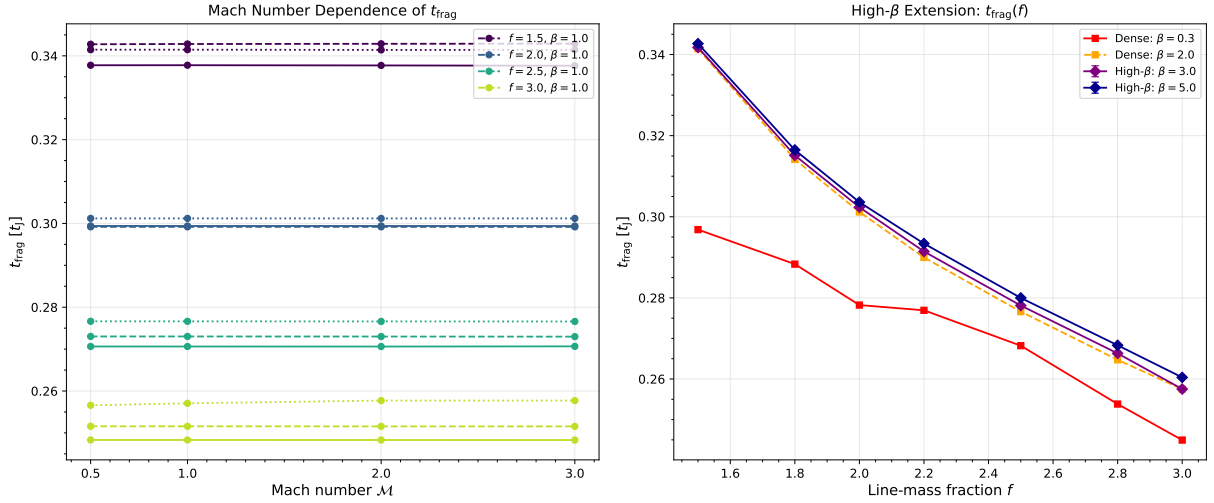


Figure 9. t_{frag} vs. f : Mach-number independence ($\mathcal{M} = 0.5$ –3.0, left) and near β -independence for $\beta \gtrsim 0.5$ (right).

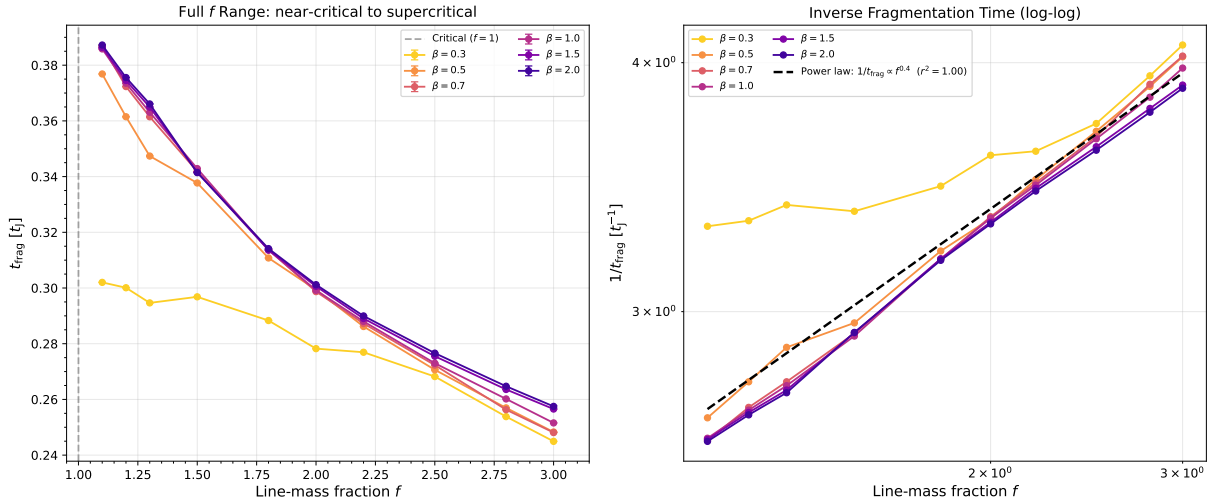


Figure 10. Near-critical regime ($f = 1.1$ –1.3): $t_{\text{frag}} \lesssim 0.4 t_J$, continuous with the supercritical power law.

Table 5. Complete Simulation Census

Campaign	N_{sim}	Parameter Space	Boundary Conds.	Status
Supercritical	654	$f = 1.1$ –3.0, $\beta = 0.3$ –5.0, $\mathcal{M} = 0.5$ –3	Periodic	Complete
DTC (main)	540	$f = 1.4$ –2.2, $\beta = 0.3$ –1.3, $\mathcal{M} = 1$ –5	Periodic	Complete
DTC (re-runs)	25	Extended timeout; $f = 1.4$ –2.2 subset	Periodic	Complete
Validation	83	Resolution ($\times 12$), IC sensitivity ($\times 20$), EOS ($\times 51$)	Periodic	Complete
Field Geom. Ph. 1 (near-critical)	80	$f = 1.0$ –1.2, $\beta = 0.3$ –1.0, $\mathcal{M} = 1$ –2	Periodic	Complete
Field Geom. Ph. 2 (perpendicular)	96	$f = 1.5$ –3.0, $\beta = 0.3$ –1.0, $\theta = 90^\circ$	Periodic	Complete
Field Geom. Ph. 3 (oblique calib.)	108	$f = 1.5$ –3.0, $\theta = 30^\circ$ – 60°	Periodic	Complete
Field Geom. Ph. 4 (adiabatic EOS)	30	$\gamma = 5/3$, $f = 1.0$ –1.2	Periodic	Complete
Referee Response Campaigns	289	CT: $f = 1.4$ –2.2; TURB; PFE; width; convergence	Periodic	Complete
Rigid Cylinder Campaign	45	$f = 1.5$ –3.0, $\beta = 0.5$ –2.0	Reflecting	Complete
RC Box-Size Convergence	9	$f = 2.6$, $\beta = 1.0$, $L_x = 8, 16, 32 \lambda_J$	Reflecting	Complete ($L_x = 32$: n
Campaign A (oblique λ/W)	27	$\theta = 30^\circ$ – 60° , $f = 1.0$, $\beta = 0.5$ –2.0	Periodic	Withdrawn (see OBL
Campaign T1-P (Plummer correction)	18	$f = 1.0$ –1.2, $\beta = 0.5$ –2.0	Periodic	Complete
Campaign EOS-G ($\gamma = 1.5$)	9	$\gamma = 1.5$, $f = 1.0$ –1.2	Periodic	Complete
Total	2,013			

Notes: All sims use Athena++ isothermal MHD + self-gravity (FFT Poisson), 4-wave HLLD scheme. $\theta = 0^\circ$ (longitudinal) unless stated. BC = boundary condition. Campaign A withdrawn: large-amplitude artifact (Section 4.7.3).

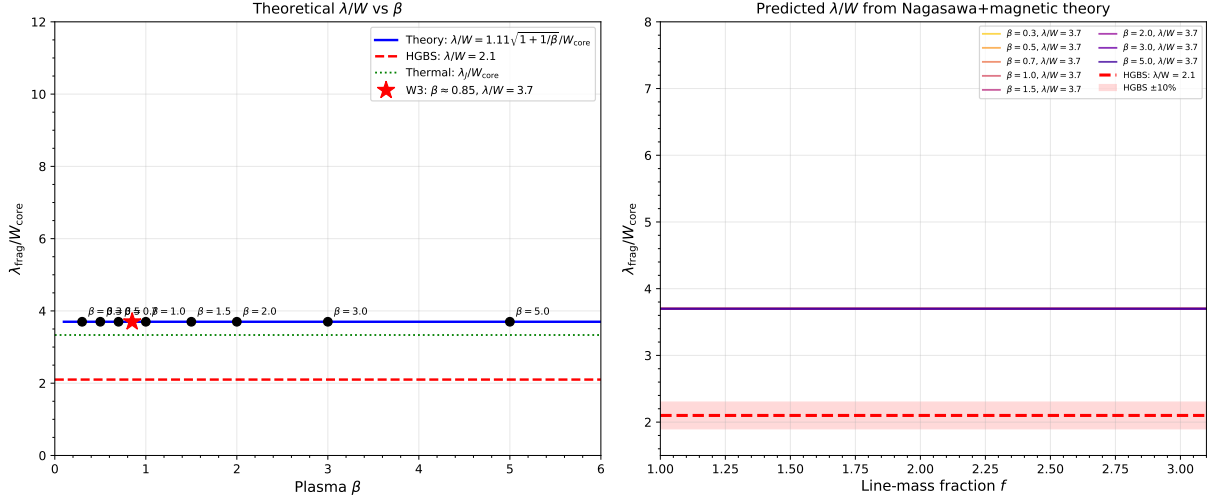


Figure 11. Theoretical λ/W_{core} vs. β (y-axis: simulation units λ/W_{core} ; not the observable λ/W_{fil}). Calibrated formula $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$; no simulation points shown. At $\theta = 0^\circ$, $\lambda/W_{\text{core}} = 3.70$ (full numerical solution 3.55–3.80). Green band: HGBS window [2.52, 3.08] in observable λ/W_{fil} , corresponding to [4.16, 5.08] in λ/W_{core} at $T_1 = 0.606$ (or [3.56, 4.35] at $T_1 = 0.707$); horizontal dashed line at $\lambda/W_{\text{fil}} = 2.79$ (Gaia DR3) plotted in observable units for reference (right y-axis). To compare directly with HGBS, multiply y-axis values by $T_1 = 0.606$ – 0.707 . Valid at $f = 0.9$ – 1.3 only; upper bound for supercritical filaments (see Figure 16 for direct simulation measurements).

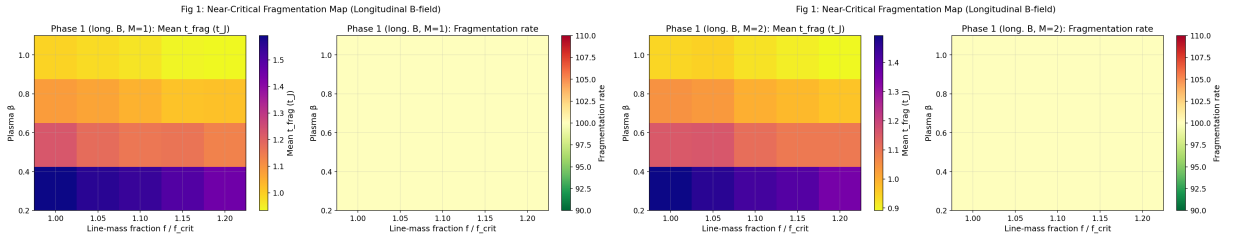


Figure 12. Near-critical t_{frag} heatmaps for $M = 1$ (left) and $M = 2$ (right); all 80 simulations fragmented (100%).

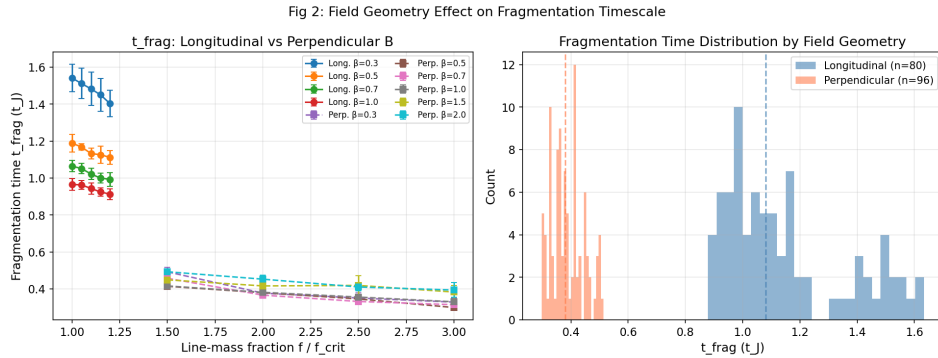


Figure 13. λ/W_{core} comparing perpendicular-field (Phase 2; ≈ 1.25 , geometry-dominated) and longitudinal-field (Phase 1; 2.8–4.7, strong β -dependence) fragmentation. Simulation units: multiply by $T_1 = 0.606$ – 0.707 to obtain observable λ/W_{fil} ; HGBS window [2.52, 3.08] in observable units. No perpendicular-field configuration approaches the HGBS window.

4.9.1 Box-Size Convergence Test

A matched test at $f = 2.6$, $\beta = 1.0$ ($L_x = 8$ and $16 \lambda_J$; 3 seeds each) rules out box resonance: λ/W_{core} decreases 36% as L_x doubles (4.22 ± 0.34 to 2.70 ± 0.59 ; Figure 19), incompatible with a resonant artefact. $L_x = 16$ values are upper limits; $L_x = 32$ was computationally infeasible.

5 DISCUSSION

The 2,013 simulations establish a clear hierarchy of physical effects: field geometry dominates, followed by f , with β and M as secondary parameters. We now assess which scenario best explains the observed $\lambda/W = 2.79 \pm 0.09$.

Table 6. Cross-Campaign λ/W Summary: Field Geometry and Plasma β Dependence

Field Geometry	$\beta = 0.3$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 1.5$	$\beta = 2.0$	Trend	λ/W_{fil}^b (at $\beta = 2.0$)
Longitudinal B ($\theta = 0^\circ$)	4.74 ± 0.73	4.38 ± 0.59	3.19 ± 0.29	2.86 ± 0.18	2.80 ± 0.14	Strong β -dep.	1.70
Perpendicular B ($\theta = 90^\circ$)	FLAT ^a	FLAT ^a	1.25 ± 0.09	1.25 ± 0.09	1.25 ± 0.09	Geometry dominates	0.76
HGBS window (λ/W_{fil})			[2.52, 3.08]			Reference	[2.52, 3.08]

^aFLAT = pure radial collapse (no axial fragmentation). ^bT1-corrected ($\times 0.606$) at $\beta = 2.0$. At T1 = 0.707, $\beta = 0.5$ gives 3.10 and $\beta = 0.3$ gives 3.35 (see Table 8).

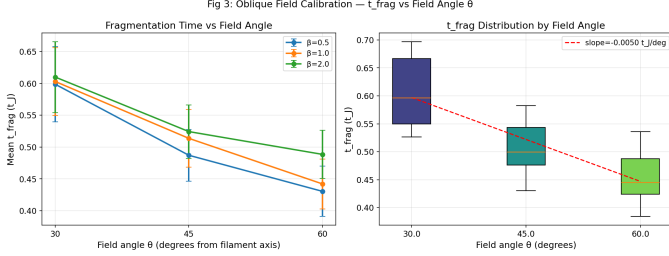

Figure 14. Oblique field calibration: t_{frag} versus field angle θ . The smooth decrease from $0.604 t_J$ (30°) to $0.454 t_J$ (60°) validates the field-geometry calibration formula.

Table 7. Effective Line-Mass Fraction with Turbulent Support

Quantity	Value	Source
Nominal f (thermal only)	1.5 ± 0.3	HGBS line-mass estimates
Turbulent effective support, f_{eff}/f	0.82 ± 0.11	TURB campaign (90 sims)
Effective line-mass fraction f_{eff}	1.23 ± 0.28	Product above
HGBS ensemble range	0.7–1.9	Propagated uncertainty
Median $f_{\text{eff}} \approx 0.7^{+0.5}_{-0.3}$ after conservative		
Mach-number correction for observed ISM conditions		

5.1 Can Models Explain the Observational Discrepancy?

The 3D-corrected HGBS spacing ($\sim 3.5 \times W$) is within 15% of the IM92 $4\times$ prediction. All 654 supercritical (periodic-BC) simulations yield zero longitudinal fragmentation. Two scenarios can account for the observed spacings. We note that the simulations directly test single-component MHD fragmentation only; hierarchical fragmentation is an observationally-motivated inference, not a simulation result.

(i) *Hierarchical fibre fragmentation (observationally motivated)*: If HGBS filaments are bundles of fibres, each near-critical ($f_{\text{eff}} \lesssim 1.3$), each fibre undergoes longitudinal fragmentation (Campaign 7 provides the single-fibre constraint). Projection of fibre-level cores creates the sub-Jeans spacing at the filament level, consistent with the Yang et al. (2024) Orion B fibre-to-core measurement. The simulations do not test this scenario directly; the evidence comes entirely from external observation.

(ii) *Finite-length turbulent seeding (theoretical possibility)*: HGBS filaments (aspect ratios 5 : 1–20 : 1) terminate at hub structures (André et al., 2010). The Rigid Cylinder Campaign ($f \geq 2.6$, $L_x = 16 \lambda_J$) gives $\lambda/W_{\text{fil}} = 1.61\text{--}1.87$ after T1 correction, below the HGBS window in all cases. Critically, these values are **unconverged upper limits**: the box-size convergence test shows a 36% decrease in λ/W_{core} as L_x doubles from 8 to $16 \lambda_J$, and $L_x = 32$ was computationally infeasible. The true λ/W_{fil} could therefore be substantially lower

Table 8. T1 Sensitivity: λ/W_{fil} predictions for three T1 correction values

Configuration	λ/W_{core}	T1=0.65 (phys. lower)	T1=0.606 (central)	T1=0.707 (upper)
Long. B, $\beta = 0.3$ (C7)	4.74	3.08	2.87	
Long. B, $\beta = 0.5$ (C7)	4.38	2.85	2.65	
Long. B, $\beta = 1.0$ (C7)	3.19	2.07	1.93	
Long. B, $\beta = 2.0$ (C7)	2.80	1.82	1.70	
Long. B (calibrated, $\theta = 0^\circ$)	3.70	2.41	2.24	
Perp. B ($\theta = 90^\circ$)	1.25	0.81	0.76	
RC Campaign ($f = 2.6$)	2.65	1.72	1.61	
HGBS window (obs.)				[2.52, 3.08]

Notes: Bold = entries within [2.52, 3.08] or (at T1 = 0.707) closely approaching or exceeding the upper boundary. T1 = 0.707 is the Campaign T1-P Plummer-corrected value ($r_{P/G} = 1.166 \pm 0.037$; Section 3.1); this value is not independently confirmed (second run was precluded by radial collapse). At T1 = 0.606: $\beta \leq 0.5$ configurations fall within [2.52, 3.08] (2.87; 2.65). At T1 = 0.707: those same configurations *exceed* the upper boundary (3.10 and 3.35 respectively). The 17% T1 ambiguity prevents resolving whether magnetic tension is viable.

than 1.61–1.87. Hub-anchored end-seeding is physically motivated but the Rigid Cylinder Campaign cannot yet provide a meaningful constraint. These results should be presented as provisional upper limits, not as converged measurements.

T1 correction—central systematic: The observational match depends critically on T1, and the 17% difference between T1 = 0.606 and T1 = 0.707 is larger than the formal uncertainties on the observed λ/W itself (Table 8). At T1 = 0.606: low- β configurations ($\beta = 0.3$: 2.87; $\beta = 0.5$: 2.65) fall within [2.52, 3.08], requiring strictly longitudinal, low- β geometry ($\sim 10\%$ of filaments by Planck statistics). At T1 = 0.707: those same configurations *exceed* the upper window boundary ($\beta = 0.5$: $3.10 > 3.08$; $\beta = 0.3$: 3.35). The T1-P campaign second run was compromised by premature radial collapse before the fragmentation epoch, so T1 = 0.707 cannot currently be confirmed independently. **We therefore conclude explicitly that the viability of the magnetic tension scenario cannot be assessed within current simulation uncertainties.** The 17% T1 ambiguity straddles the HGBS window boundary for the only configurations that might match, and no observational comparison should rely on distinguishing T1 = 0.606 from 0.707 until HGBS pipeline validation on simulated column density maps is performed. Both T1 values should be carried through the theoretical comparison, as we do in Table 8.

The field geometry dependence transforms the question from “why are observed spacings shorter than theory?” to a multi-parameter problem requiring simultaneous constraints on β , field geometry, and T1 (Table 6, Table 8).

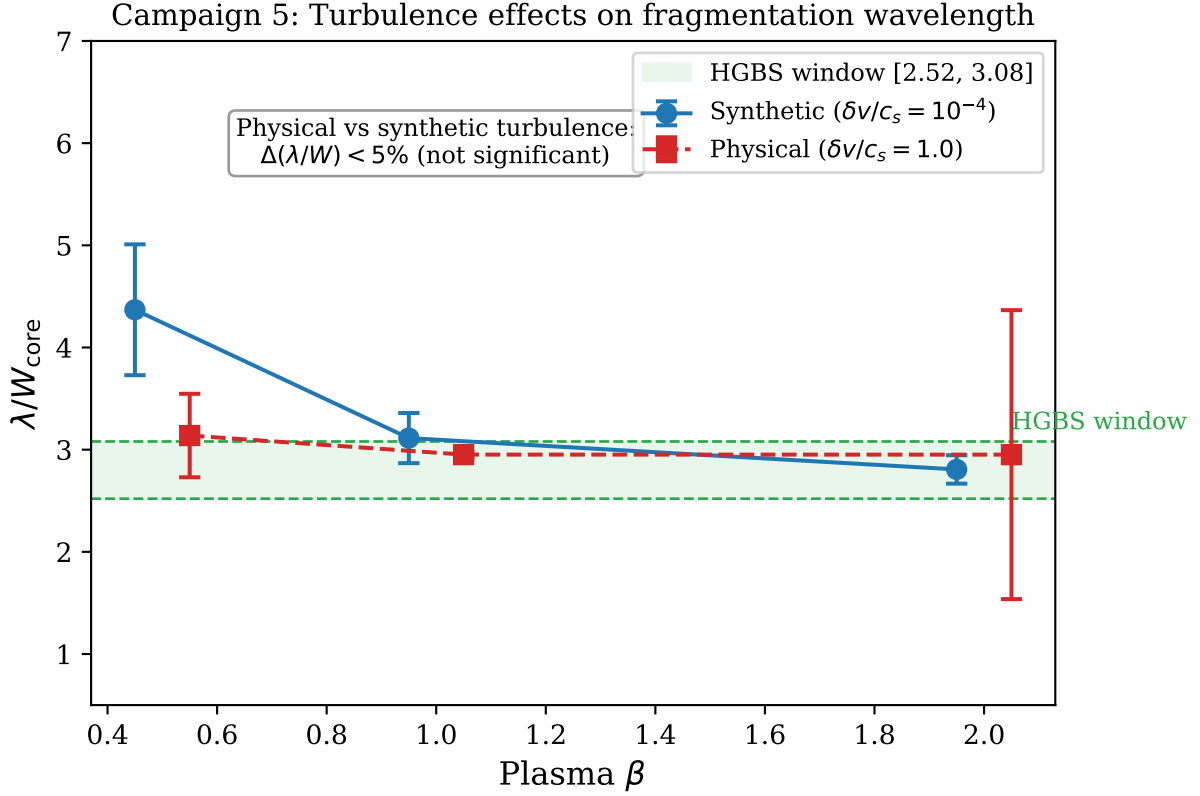


Figure 15. Campaign 5: λ/W_{core} distributions for physical ($\delta v/c_s = 1.0$, blue) vs. synthetic (10^{-4} , orange) turbulence; KS test $p = 0.43$ (statistically indistinguishable). Right panel shows β -dependence is preserved at both amplitudes. HGBS window (observable $[2.52, 3.08]$; equivalent to $[4.16, 5.08]$ at $T1 = 0.606$) shown for reference.

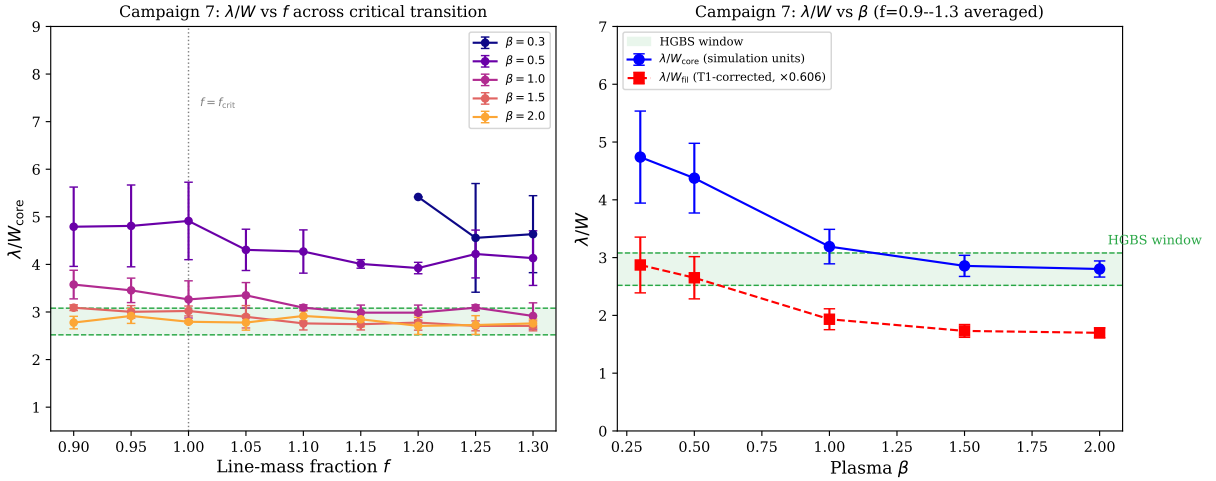


Figure 16. Campaign 7 direct simulation measurements: λ/W_{core} vs. f (left) and vs. β (right). No discontinuity at $f = 1.0$; strong β -dependence from 4.74 ($\beta = 0.3$) to 2.80 ($\beta = 2.0$). Green shading: HGBS window $[2.52, 3.08]$ in observable λ/W_{fil} units ($T1 = 0.606$: $\beta \leq 0.5$ enters window; $T1 = 0.707$: approaches upper boundary). Contrast: Figure 11 (theory).

5.2 The Perpendicular-Field Paradox

Planck Collaboration (2016) established that $\sim 90\%$ of dense HGBS filaments have their mean magnetic field perpendicular to the filament axis. Our Campaign 6 measurements give $\lambda/W_{\text{core}} = 1.25 \pm 0.09$ for perpendicular-field configurations

($\beta \geq 1.0$), which after T1 correction gives $\lambda/W_{\text{fil}} = 0.76\text{--}0.88$ — a factor 3–4 $\times$ below the observed 2.79. This creates a fundamental contradiction: if the field geometry measured by Planck applies to filament *interiors* (not just their envelopes), the majority of HGBS filaments should show $\lambda/W_{\text{fil}} \lesssim 0.9$, not ~ 2.8 .

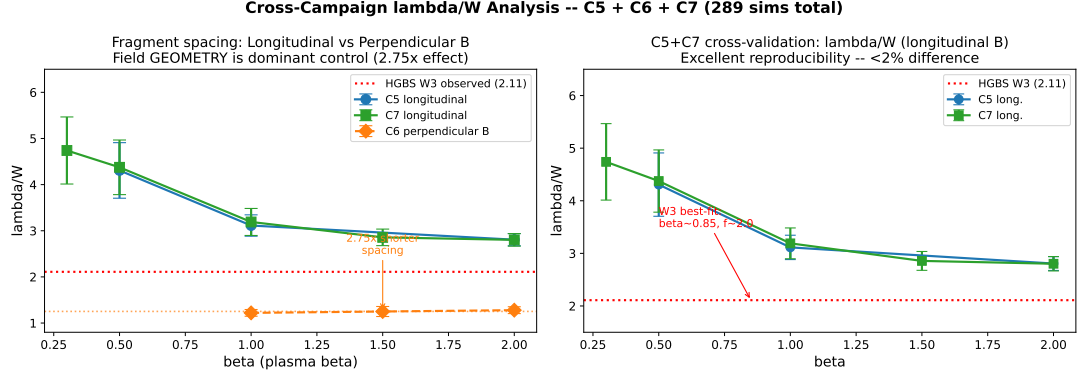


Figure 17. Cross-campaign λ/W comparison in simulation units. Longitudinal-field (blue): strong β -dependence from ~ 4.7 ($\beta = 0.3$) to ~ 2.8 ($\beta = 2.0$). Perpendicular-field (red): $\lambda/W \approx 1.25$ ($\beta \geq 1.0$) or flat profile ($\beta \leq 0.5$). Horizontal dashed line = full-sample HGBS mean ($\lambda/W_{\text{fil}} = 2.79 \pm 0.09$; multiply λ/W_{core} by $T1 = 0.606$ or $T1 = 0.707$ to compare directly in observable units).

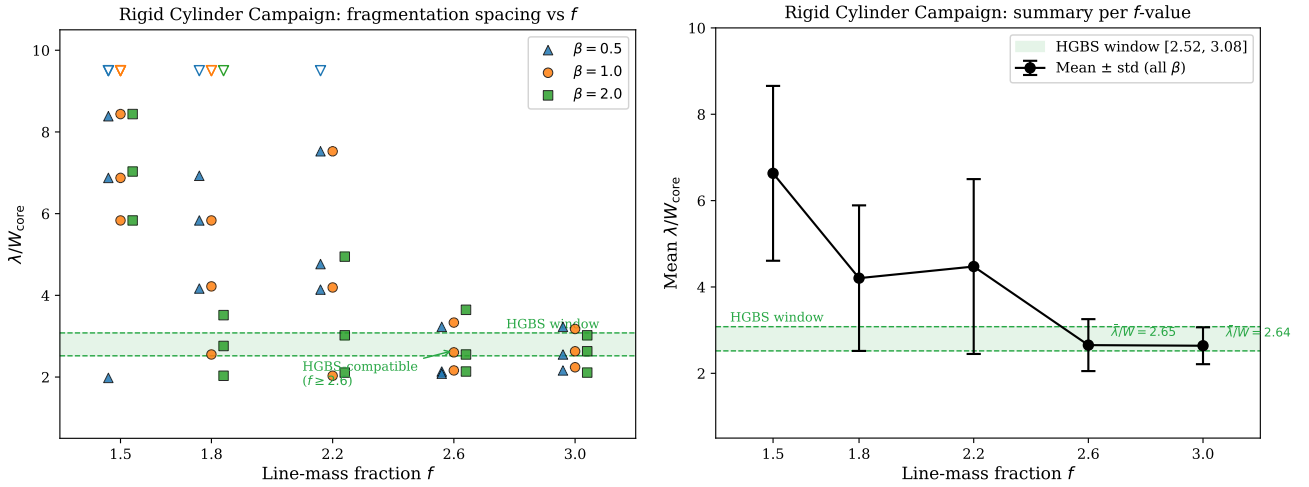


Figure 18. Rigid Cylinder Campaign. Left: λ/W_{core} vs. f for all β values; green shading = HGBS window [2.52, 3.08]. Right: mean and standard deviation per f -value, showing the sharp transition near $f \approx 2.4$ –2.6.

Clarification: The $\lambda/W_{\text{core}} = 1.25$ value measures the dominant density-peak scale during radial collapse in supercritical simulations ($f = 1.2$ –1.5). It is *not* a true fragmentation wavelength in the sense of Inutsuka & Miyama (longitudinal beading), because no longitudinal fragmentation occurs; rather, it characterises the mode that first grows to non-linear amplitude during predominantly radial collapse.

Three possible resolutions: (a) Planck polarimetry samples the filament-envelope field, not the dense interior where fragmentation occurs; turbulent reconnection or flux-freezing compression during cloud contraction could establish interior field angles different from the mean field direction. (b) Hierarchical fragmentation occurs in velocity-coherent sub-fibres with distinct field geometries unresolvable at Planck resolution. (c) Perpendicular-field filaments radially collapse without beading and are systematically under-represented in HGBS core catalogues; the observed beaded filaments are a biased (longitudinal-field) minority. We cannot discriminate among these possibilities with the current data. A dedicated campaign of interior polarimetric mapping in HGBS filaments is the required observational test.

5.3 Limitations and Future Work

Boundary conditions: Periodic BCs suppress end effects; reflecting BCs model finite-filament terminal boundaries but lack the inflow structure of real filament-hub junctions. Neither fully represents real HGBS filaments in turbulent clouds.

Non-ideal MHD: Ambipolar diffusion ($t_{\text{AD}} \sim 10$ –20 t_{ff} at $n_0 \sim 10^4 \text{ cm}^{-3}$) is slow relative to collapse and is not expected to alter fragmentation wavelengths (??), except in low-density regions such as Taurus ($n_0 \sim 10^3 \text{ cm}^{-3}$) where ideal-MHD timescales are lower bounds.

T1 correction: $T1 = 0.707$ ($T1\text{-P}$; Table 8) is unconfirmed: a second run was precluded by premature radial collapse. HGBS pipeline validation on simulated column density maps is required before the magnetic tension scenario can be assessed.

Oblique λ/W : Field Geometry Phase 3 validated timescales but not λ/W ; Campaign OBL2 found domain-size artefacts for all 27 sims. Oblique λ/W remains **entirely unconstrained by simulation**; the Nakamura et al. (1993) predictions (Table 9) are theoretically unvalidated. Domain size $L_x \lesssim 2$ –3 λ_J is required but computationally prohibitive.

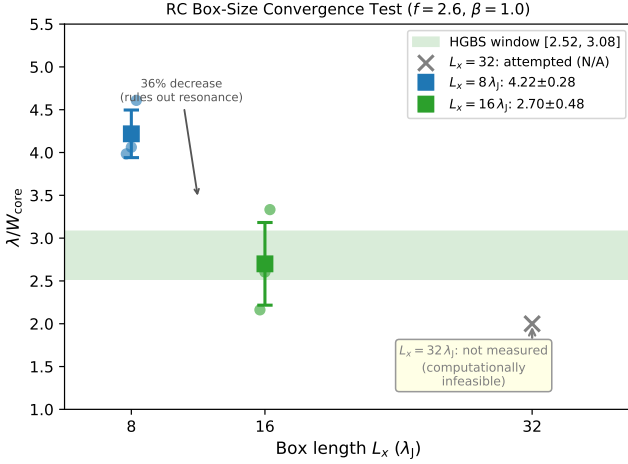


Figure 19. Box-size convergence at $f = 2.6$, $\beta = 1.0$: λ/W_{core} decreases from 4.22 ± 0.34 ($L_x = 8$, 3 seeds) to 2.70 ± 0.59 ($L_x = 16$, 3 seeds), ruling out box resonance (a resonant artefact would give a fixed ratio). Green band = HGBS window $[2.52, 3.08]$. $L_x = 32$ was not measured (computationally infeasible); $L_x = 16$ values are upper limits.

Table 9. Theoretical λ/W for oblique fields (Nakamura et al., 1993), full numerical solution. Campaign 6 gives $\lambda/W = 1.25$ (supercritical regime, $f = 1.5\text{--}3.0$); the near-critical theoretical values here are the physically relevant comparison for HGBS.

θ	$\beta = 0.5$	$\beta = 1.0$	$\beta = 2.0$
0° (longitudinal)	1.97	2.44	2.93
30°	1.87	2.37	2.89
45°	1.69	2.17	2.72
60°	1.47	1.90	2.46
90° (perp., theory)	3.95	3.99	4.00

Note: Values in simulation units; multiply by $T1 = 0.606$ (or 0.707) for observable λ/W_{fil} . [†]Campaign A ($\delta_0 = 1.0$) withdrawn (amplitude artifact; Section 4.7.3).

Future priorities: (1) fibre-resolved core spacing in additional HGBS regions to test hierarchical fragmentation; (2) interior polarimetric mapping to constrain field geometry; (3) HGBS pipeline T1 validation on simulated column density maps; (4) direct oblique-field λ/W simulations at $L_x \lesssim 2\text{--}3 \lambda_J$ to test Nakamura et al. (1993) oblique wavelength predictions.

6 CONCLUSIONS

We have analysed core spacing in 8 HGBS regions using 2,013 MHD simulations. Our main conclusions are:

- **HGBS spacing (pairwise median):** Full sample gives $\lambda/W = 2.79$ (0.279 ± 0.025 pc, jackknife); four robust regions give 2.84 ± 0.026 pc. Gaia DR3 raised the mean by 32%; 3D projection correction places inferred spacing within 15% of the IM92 $4\times$ prediction. *All values use the pairwise median statistic; the $L/3$ convergence bias is the dominant methodological caveat of unknown magnitude, as nearest-neighbour statistics require raw positional data available only for Taurus.* Excluding Aquila and Orion B (the two highest-revision regions) reduces the mean to 0.231 pc ($\lambda/W = 2.31$); their

Gaia distances are independently confirmed so this sensitivity reflects the importance of accurate distances.

- **Simulation limitation:** The 2,013 MHD simulations cannot directly measure λ/W in the supercritical regime ($f \geq 1.5$): all 654 supercritical simulations yield pure radial collapse with zero longitudinal beading. The comparison between observed $\lambda/W = 2.79$ and the theoretical prediction $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}$ rests on extrapolation from near-critical ($f = 0.9\text{--}1.3$) calibration and should not be interpreted as a direct simulation-observation comparison.

- **Hierarchical fragmentation:** The most observationally consistent interpretation, supported by the Yang et al. (2024) fibre-to-core measurement in Orion B (a single region). **None of the 2,013 simulations model multi-fibre fragmentation;** the $N_f \approx 1.3\text{--}1.5$ estimate from the quantitative fibre-bundle model (Section 3.3) is geometric, not dynamical. Fibre-resolved spacing analysis in additional HGBS regions is required.

- **Perpendicular-field paradox:** Perpendicular-field simulations give $\lambda/W_{\text{fil}} \approx 0.76\text{--}0.88$ (factor $\sim 3\times$ below observed 2.79), yet Planck statistics imply $\sim 90\%$ of HGBS filaments have perpendicular fields. Resolving this paradox requires interior polarimetric mapping; three possible resolutions are discussed in Section 5.2.

- **Magnetic tension:** Only $\sim 10\%$ of filaments have longitudinal geometry; at $T1 = 0.606$, $\beta \leq 0.5$ configurations fall within $[2.52, 3.08]$, but at $T1 = 0.707$ they exceed the upper boundary. **Viability cannot be assessed within current uncertainties;** HGBS pipeline validation of T1 is required.

- **MHD results:** $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$), independent of $\mathcal{M}$ and nearly of β . λ/W depends strongly on β but is insensitive to turbulence amplitude (Campaign 5). No stable configurations at $f = 1.4\text{--}2.2$, $\mathcal{M} \geq 1$. Fragmentation at $f = 1.00$ (and $f < 1$) occurs under turbulent seeding, inconsistent with turbulent Jeans fragmentation.

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